

Assignment 1

1. Let $A \neq \emptyset$ be bounded below.

(a) Prove that for any $\epsilon > 0$ there exists $a \in A$ such that $a < \epsilon + \inf A$.

Proof. (by contradiction) Suppose $\exists \epsilon > 0$ s.t. $\forall a \in A, \epsilon + \inf\{A\} \leq a$. Therefore $\epsilon + \inf\{A\}$ is a lower bound for A . However, $\epsilon + \inf\{A\} > \inf\{A\}$, making $\epsilon + \inf\{A\}$ a greater lower bound for A than $\inf\{A\}$. Thus, $\inf\{A\}$ is not the greatest lower bound for A . This is clearly a contradiction, thus, for all $\epsilon > 0$, $\exists a \in A$ such that $a < \epsilon + \inf\{A\}$. $\square$

(b) Prove that there is a sequence $a_n \in A$ such that $\lim_{n \rightarrow \infty} a_n = \inf A$.

Proof. Let $A \subseteq \mathbb{R}$, $\epsilon > 0$. We proceed by cases:

C_1 : Suppose $\inf\{A\} \in A$. We may pick the sequence $a_n = \inf\{A\} \forall n \in \mathbb{N}$.

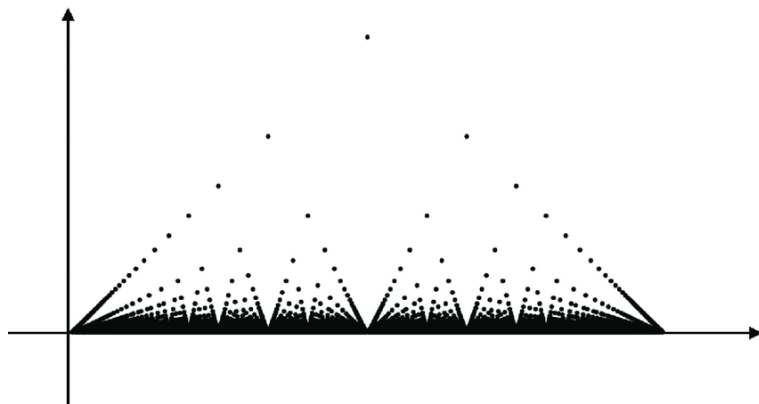
Clearly, $\lim_{n \rightarrow \infty} a_n = \inf\{A\}$.

C_2 : Suppose $\inf\{A\} \notin A$. Consider the intervals $I_n = (\inf\{A\}, \inf\{A\} + \frac{1}{n})$.

By part (a), there must exist $a_n \in I_n$ for all I_n . Thus we construct our sequence $\{a_n\}$, with each $a_n \in I_n$. For any $\epsilon > 0$, we can pick N sufficiently large such that $\frac{1}{N} < \epsilon$ (by the Archimedean Property). Thus, $\forall n > N, |\inf\{A\} - a_n| < \epsilon$. Thus, $\lim_{n \rightarrow \infty} a_n = \inf\{A\}$. $\square$

2. We define the function $\phi : (0, 1) \rightarrow \mathbb{R}$ by

$$\phi(x) = \begin{cases} 0, & \text{if } x \text{ is irrational} \\ \frac{1}{q}, & \text{if } x = \frac{p}{q} \text{ with } p \in \mathbb{Z}, q \in \mathbb{N} \text{ coprime.} \end{cases}$$



(a) Prove that if x is rational then ϕ is not continuous at x .

Proof. Let $x \in \mathbb{Q}$. Let $\{x_n\}$ be a sequence of irrational numbers such that $\lim_{n \rightarrow \infty} x_n = x$ by density of $\mathbb{Q}$ in $\mathbb{R}$. Clearly, by definition of ϕ , $\forall n \in \mathbb{N}$, $\phi(x_n) = 0$ as each x_n is irrational. Thus, $\lim_{n \rightarrow \infty} \phi(x_n) = 0$. However, $\phi(\lim_{n \rightarrow \infty} x_n) = \phi(x) = \frac{1}{q}$ for some $q \in \mathbb{N}$. Because $\lim_{n \rightarrow \infty} \phi(x_n) \neq \phi(\lim_{n \rightarrow \infty} x_n)$, ϕ is not continuous at x . Because x was chosen arbitrarily, this ϕ is not continuous at all rational numbers. $\square$

(b) Prove that if x is irrational then ϕ is continuous at x . (This is fairly challenging)

Proof. Let $\{x_n\}$ be a sequence in $(0, 1)$ s.t. $\{x_n\} \rightarrow x \in \mathbb{Q}^C$. We want to show that the $\phi(x_n) \rightarrow 0$. Notice that $x - \frac{1}{10^n} < \frac{\lfloor x \cdot 10^n \rfloor}{10^n} < x < \frac{\lfloor x \cdot 10^n \rfloor + 1}{10^n} < x + \frac{1}{10^n}$. Thus, $\frac{\lfloor x \cdot 10^n \rfloor}{10^n} < \frac{p}{q} = x_n < \frac{\lfloor x \cdot 10^n \rfloor + 1}{10^n}$. And so we define $\rho(n)$ s.t. if $x \cdot 10^n$ is

even, then $\rho(n) = 0$, otherwise $\rho(n) = 1$. Therefore $\frac{\lfloor x \cdot 10^n \rfloor - \rho(n)}{10^n} < \frac{p - q\rho(n)}{q} < \frac{\lfloor x \cdot 10^n \rfloor - \rho(n)}{10^n}$. This allows q to equal 10^n , as p must now be odd. Therefore p and q are coprime. Additionally, $\phi(\frac{p}{q} = x_n) = \frac{1}{q} = \frac{1}{10^n}$. Therefore, by the Archimedean property, as $n \rightarrow \infty$, $\phi(x_n) \rightarrow 0$. Because $x_n \rightarrow x \in \mathbb{Q}^C \Rightarrow \phi(x_n) \rightarrow \phi(x)$, ϕ is continuous for all $x \in \mathbb{Q}^C$. $\square$

Hint: Use the density of the rational and irrational numbers.

3. Let $A, B \subseteq \mathbb{R}$ with $|B| = 0$. Prove that $|A \cup B| = |A|$.

Proof. Let $A, B \subseteq \mathbb{R}$, and let $|B| = 0$. By the property of countable subadditivity, we know that $|A \cup B| \leq |A| + |B| = |A|$. It is also true that $A \subseteq (A \cup B)$, thus by the ordering property, $|A| \leq |A \cup B|$. Finally, we have that:

$$|A \cup B| \leq |A| \leq |A \cup B|.$$

Therefore $|A \cup B| = |A|$. □

4. Prove that $|\mathbb{Q}^C \cap [0, 1]| = 1$.

Proof. We have just shown that if $A, B \subseteq \mathbb{R}$ and $|B| = 0$, then $|A \cup B| = |A|$. Consider $(\mathbb{Q} \cap [0, 1]) \cup (\mathbb{Q}^C \cap [0, 1]) = [0, 1]$. The set $\mathbb{Q} \cap [0, 1]$ is countable, thus $|\mathbb{Q} \cap [0, 1]| = 0$. Thus, by the previous result,

$$1 = |[0, 1]| = |(\mathbb{Q} \cap [0, 1]) \cup (\mathbb{Q}^C \cap [0, 1])| = |\mathbb{Q}^C \cap [0, 1]|.$$

Therefore $|\mathbb{Q}^C \cap [0, 1]| = 1$. □